

# Deltares



## Sea level monitor

Corrections for atmospheric and tidal effects

# Outline

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**01** A Global Tide and Storm-surge Model (GTSM)

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**02** Surge correction and Sea Level Monitor /  
Zeespiegelmonitor

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**03** Nodal Tides

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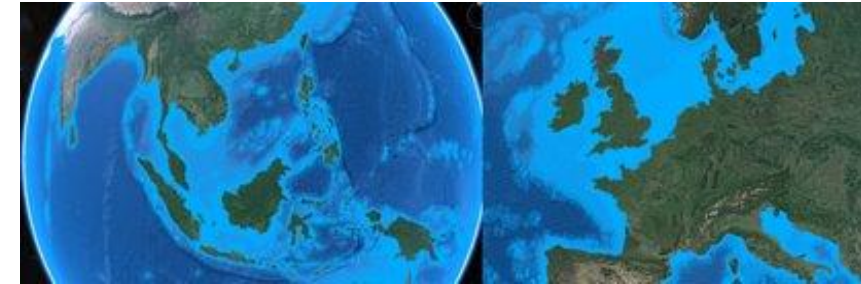
**04** Conclusions & outlook

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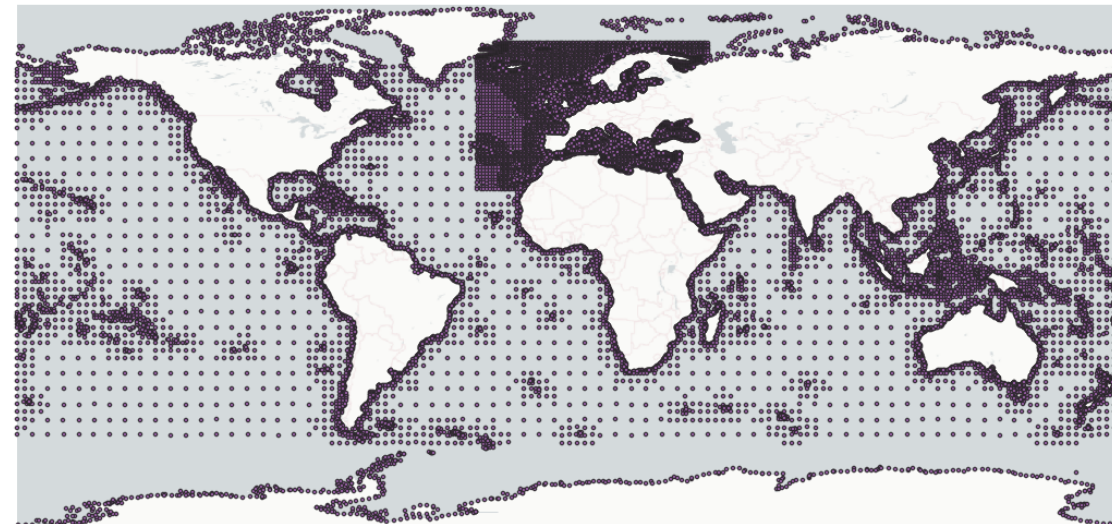
# **01 A Global Tide and Storm-surge Model (GTSM)**

## Global Tide and Storm surge Model

- **Global, depth-averaged (barotropic) hydrodynamic model** for tides, storm surges, and total water levels
- **Based on Delft3D Flexible Mesh**, using an unstructured grid with variable resolution
- **High coastal resolution** (km-scale at the coasts; coarser offshore)
- **Physically-based forcing:** tides, wind stress, atmospheric pressure, bottom friction, and tide–surge interaction
- **Self-contained global domain**, no lateral open boundaries
- **Broad applications:** operational forecasting (GLOSSIS), historical reanalysis, and climate-change impact & coastal flood-risk assessments



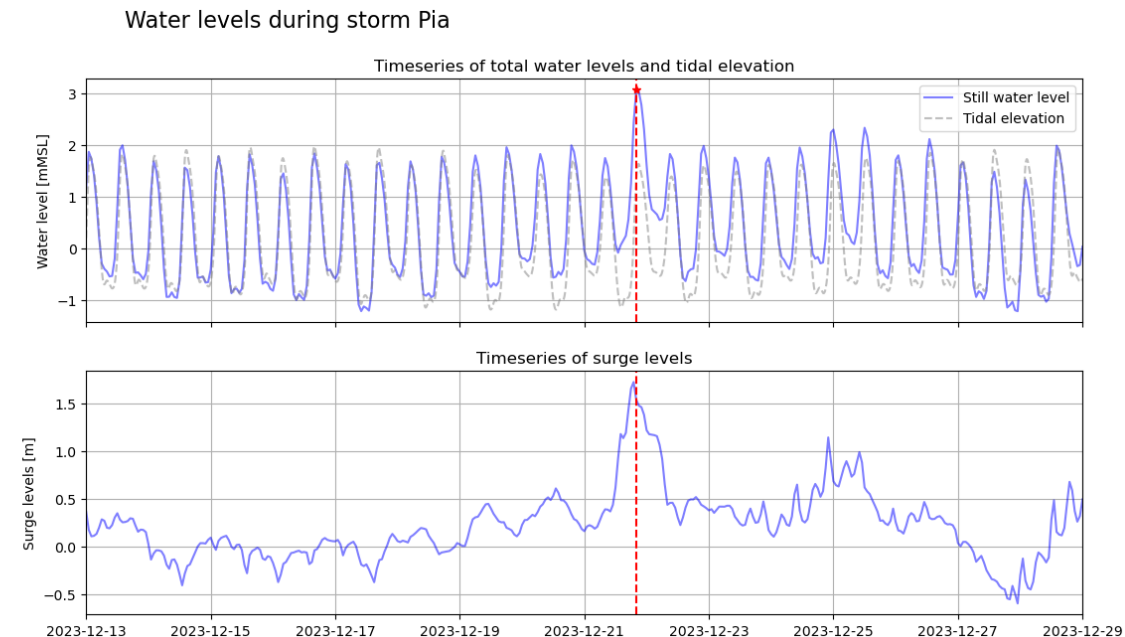
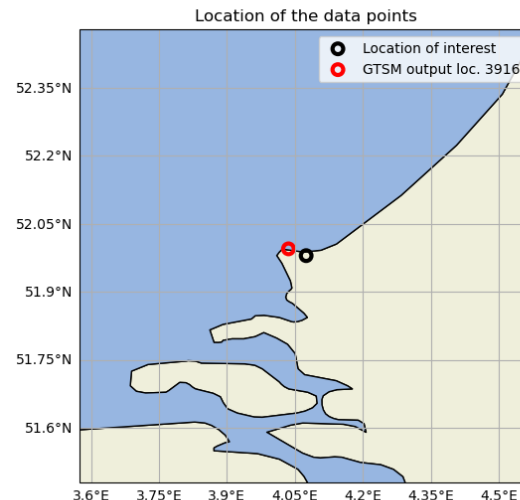
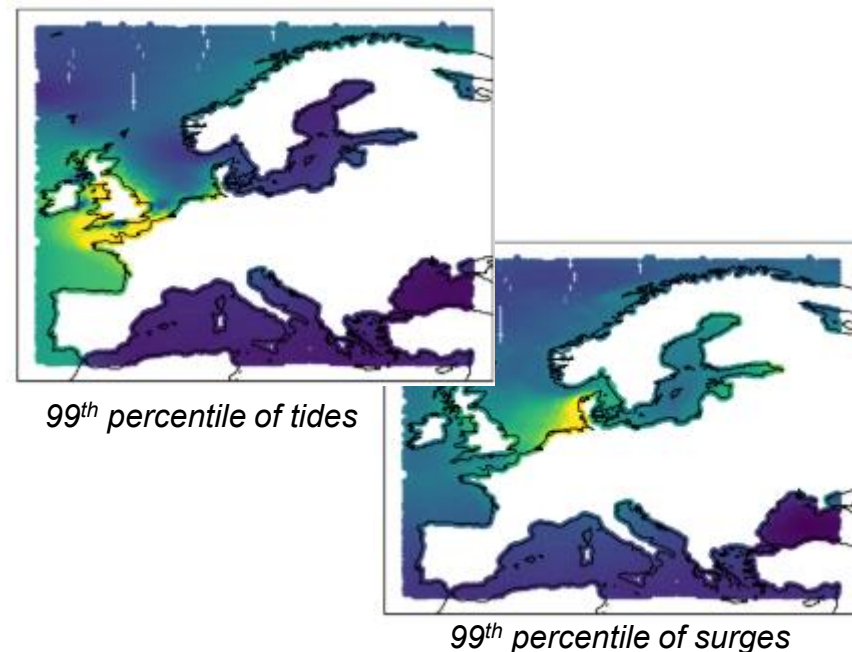
*Model grid of GTSM in Southeast Asia and Europe*





## Global Tide and Storm surge Model

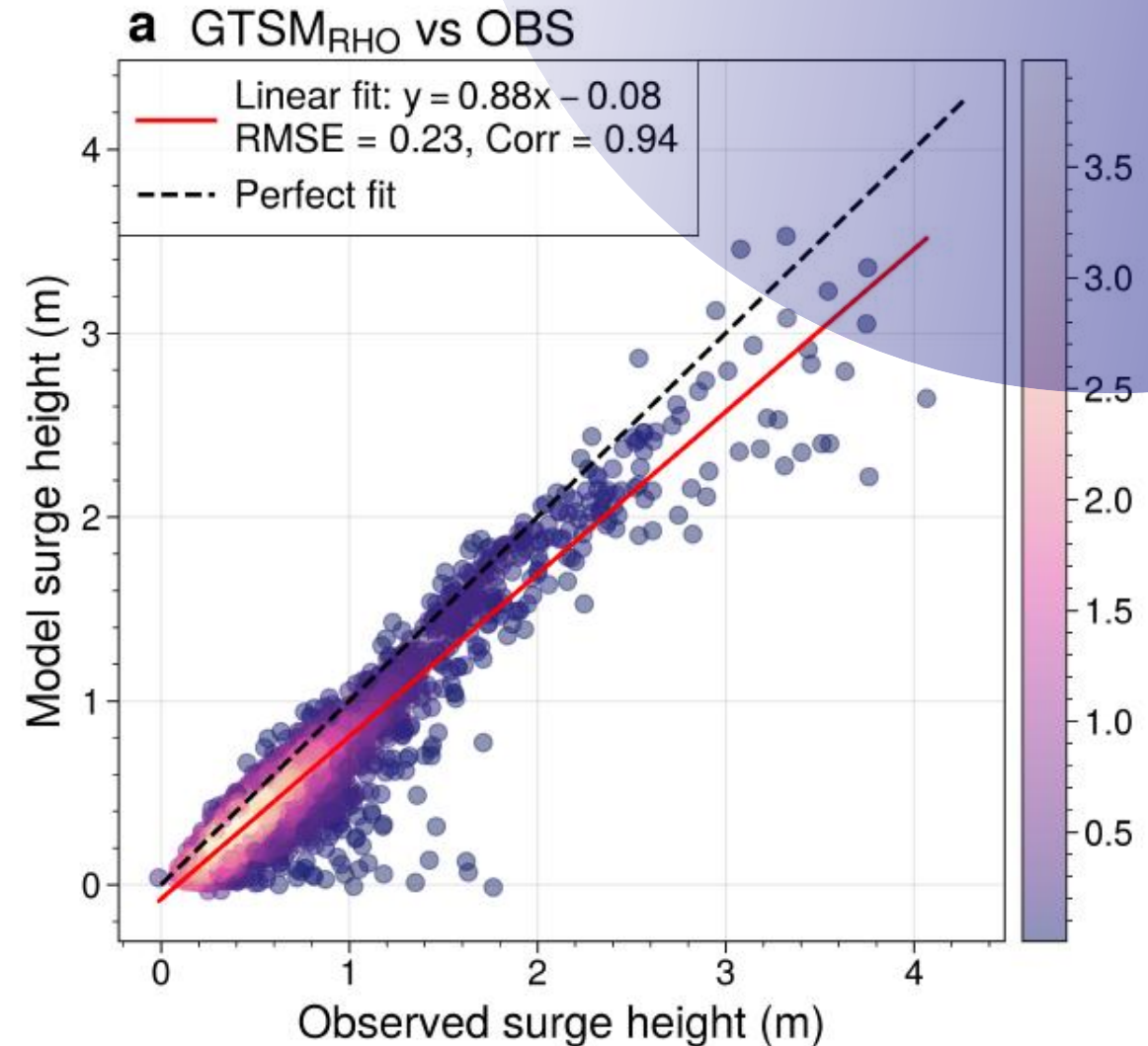
- **GTSM-ERA5 reanalysis** of storm surges is available for 1950-2025.
  - Values at Copernicus Climate Data Store include sea-level rise (although this cancels)
  - Values for Sea Level monitor do not include sea-level rise
- **Storm surge timeseries** derived as a difference between tide-surge and tide-only GTSM simulations



# Surge validation

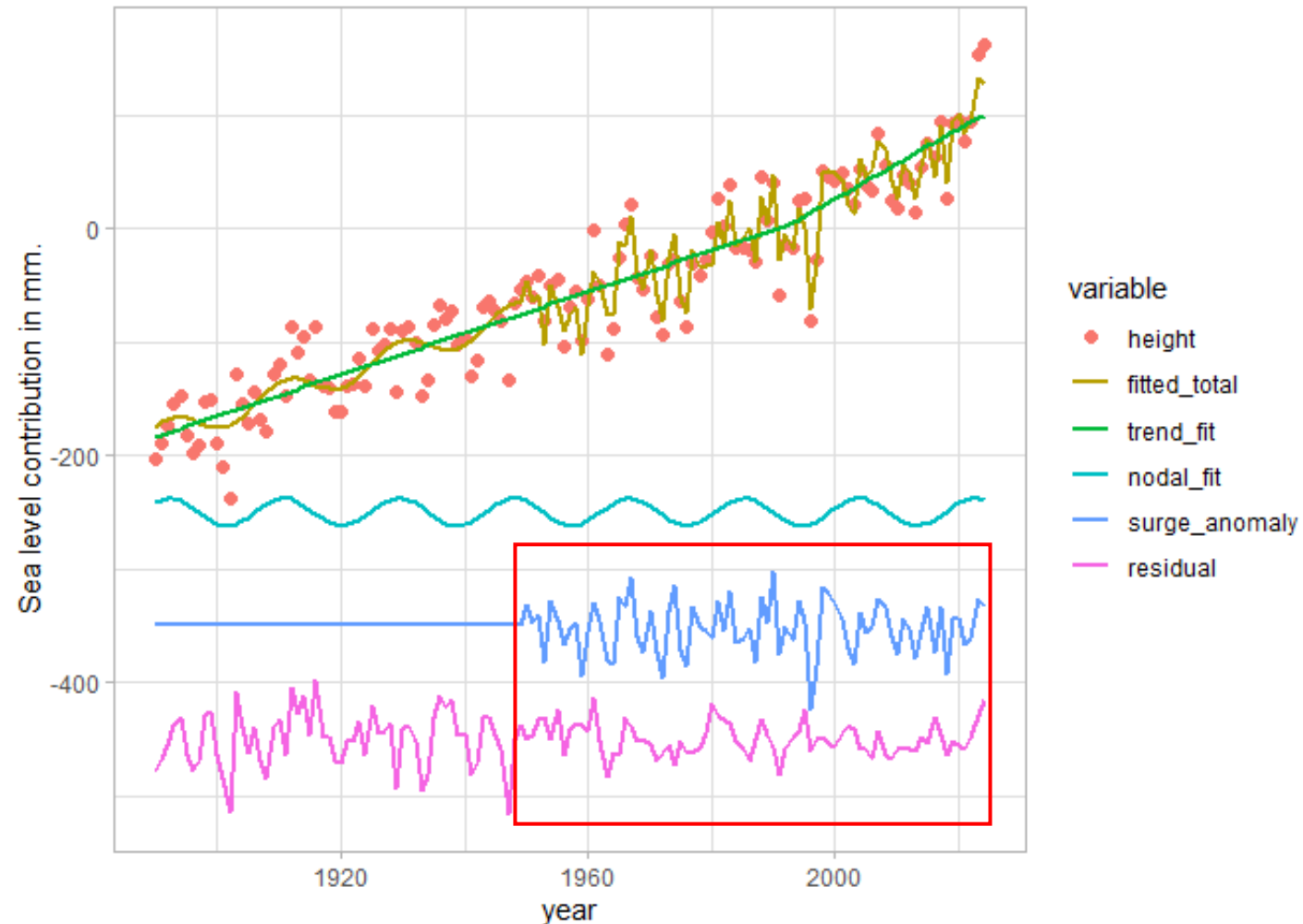
Comparison GTSM annual max surge to observations for 300 stations worldwide for 2006-2015

Nur Ozkan – under review



# 02 Surge correction and Sea Level monitor

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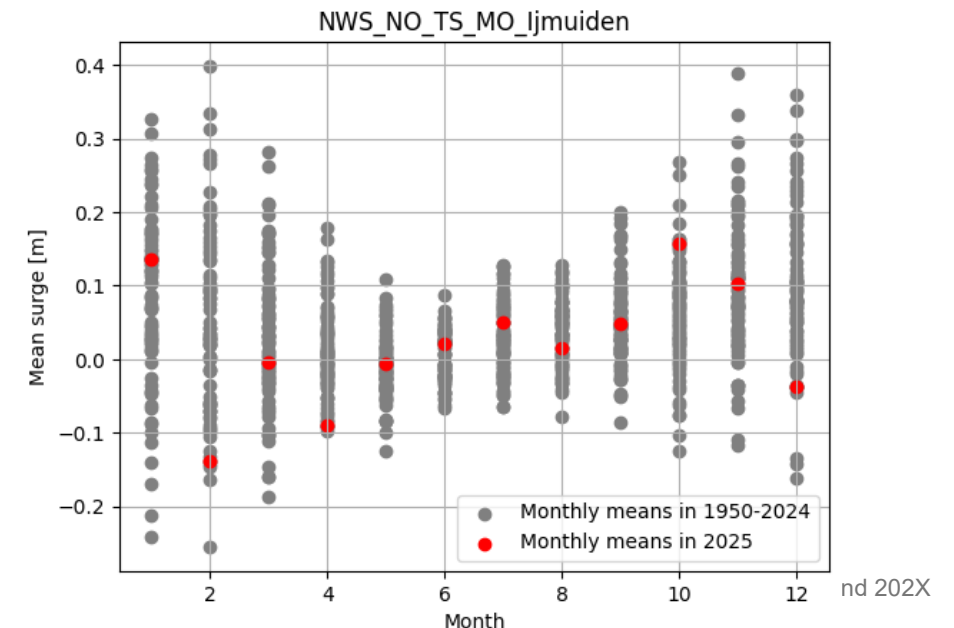
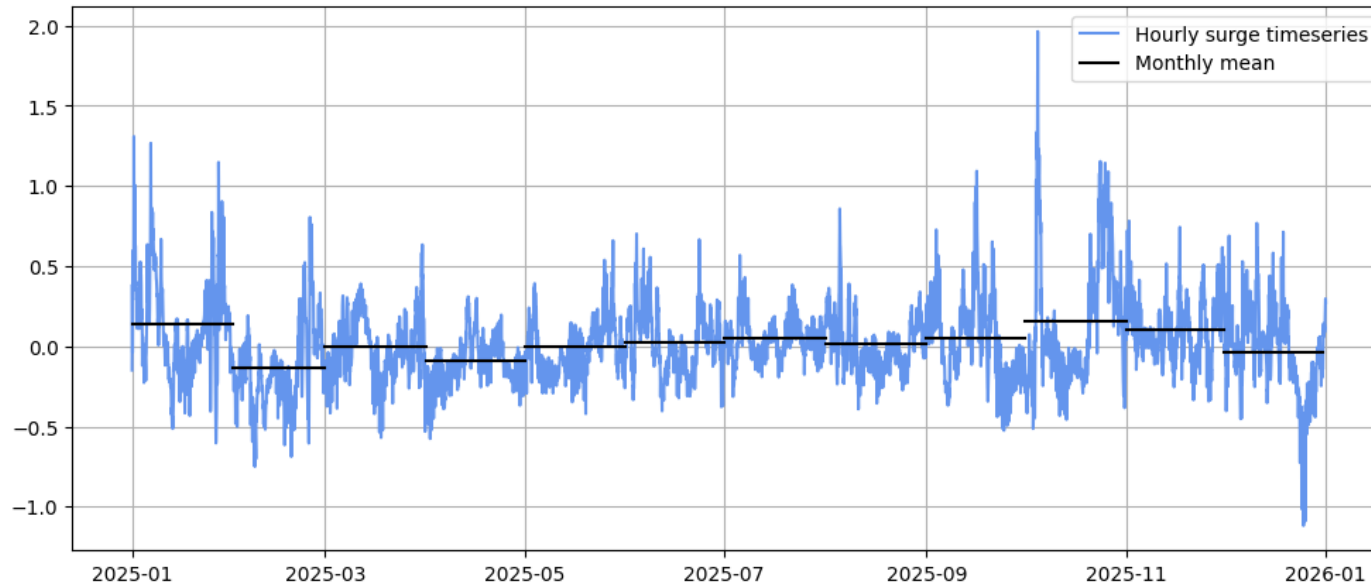


# Surge correction and Sea Level monitor

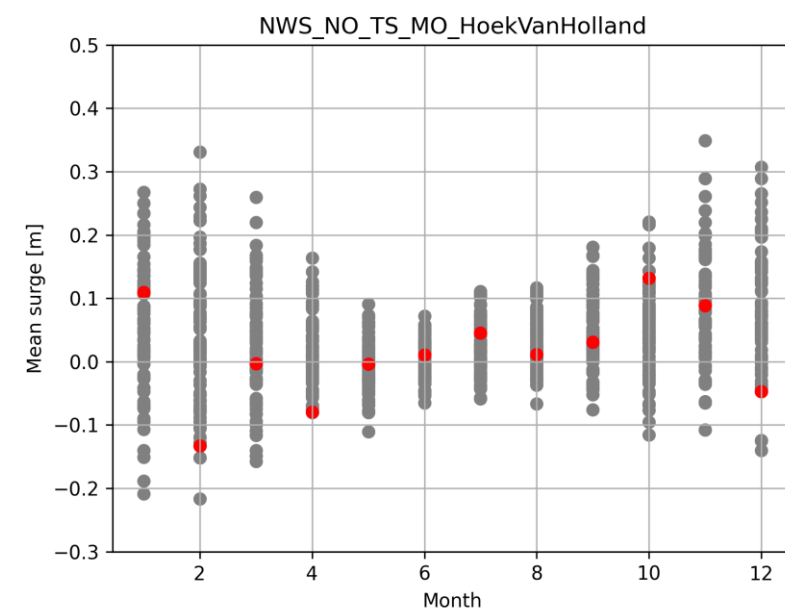
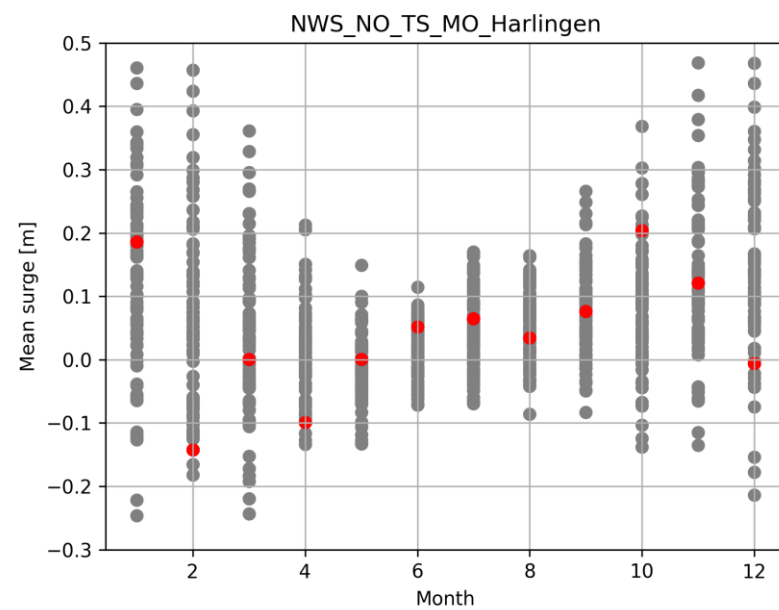
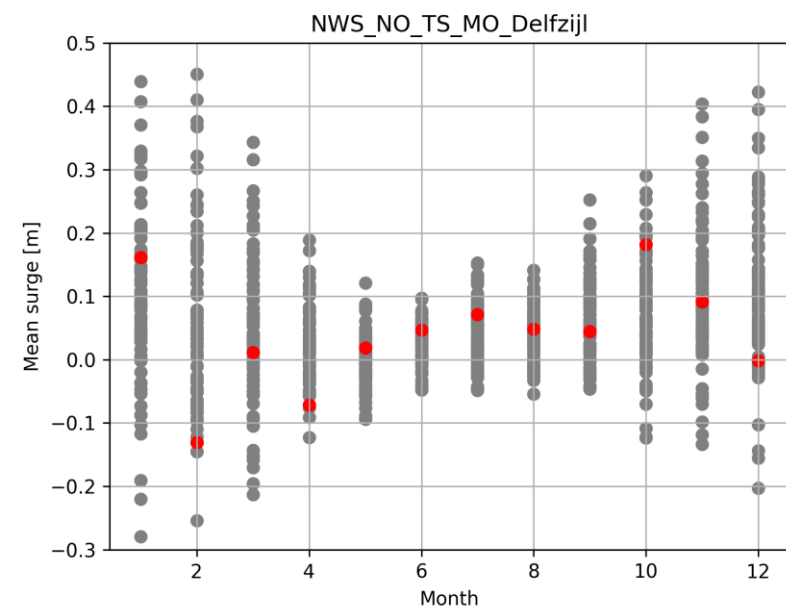
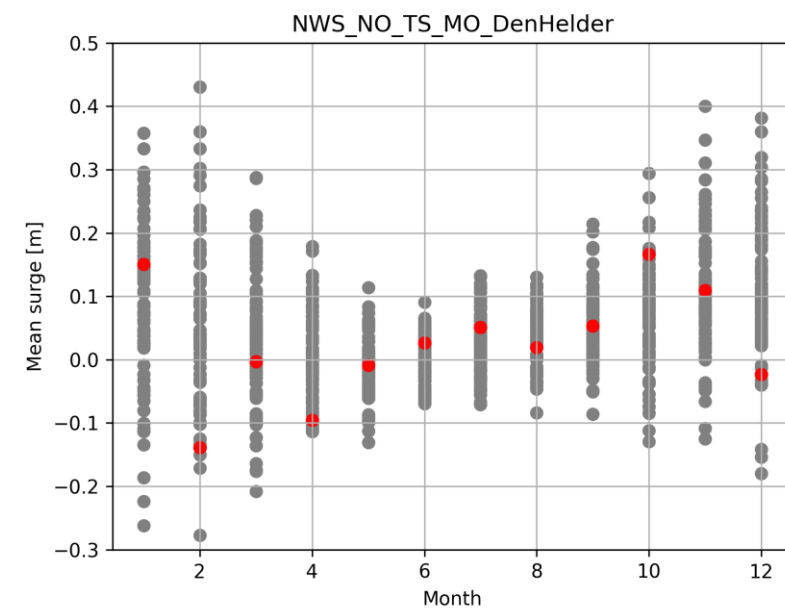
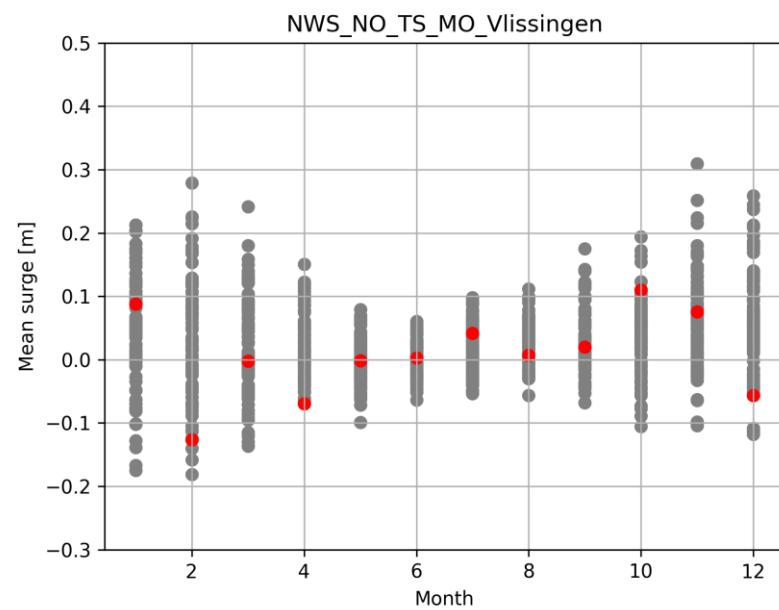
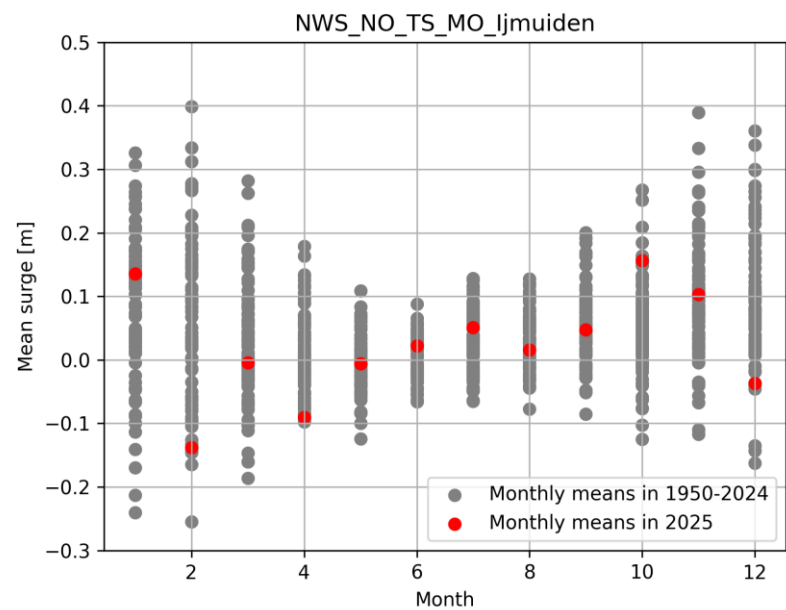
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- Annual surge level means from GTSM-ERA5 are used to correct for atmospheric effects
- Monthly means are first derived from the surge hourly timeseries, then averaged to annual values
- The magnitude of monthly means is between -30 to 50 cm, with annual means in the range -3 cm to 14 cm

Monthly mean surge at NWS\_NO\_TS\_MO\_Ijmuiden in 2025

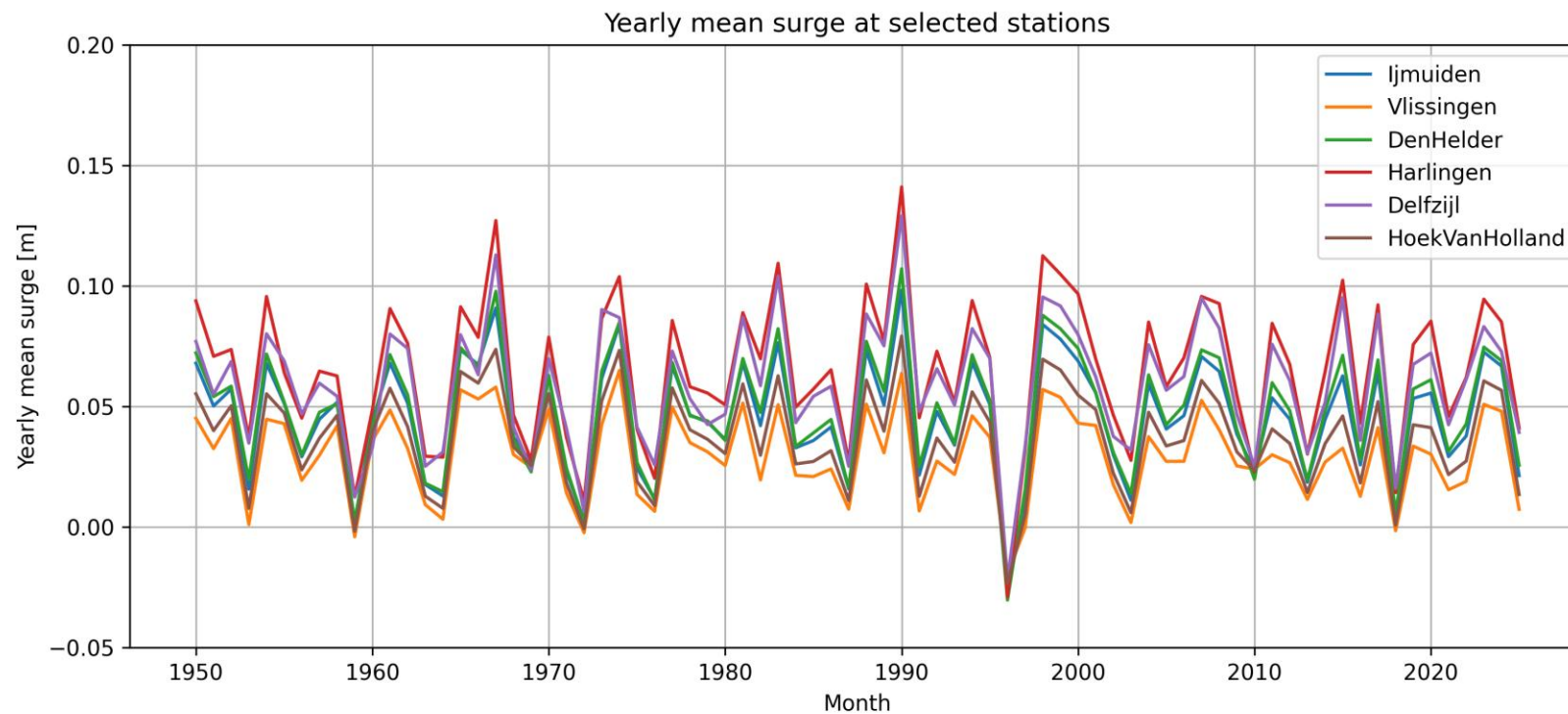






# Surge correction and Sea Level monitor

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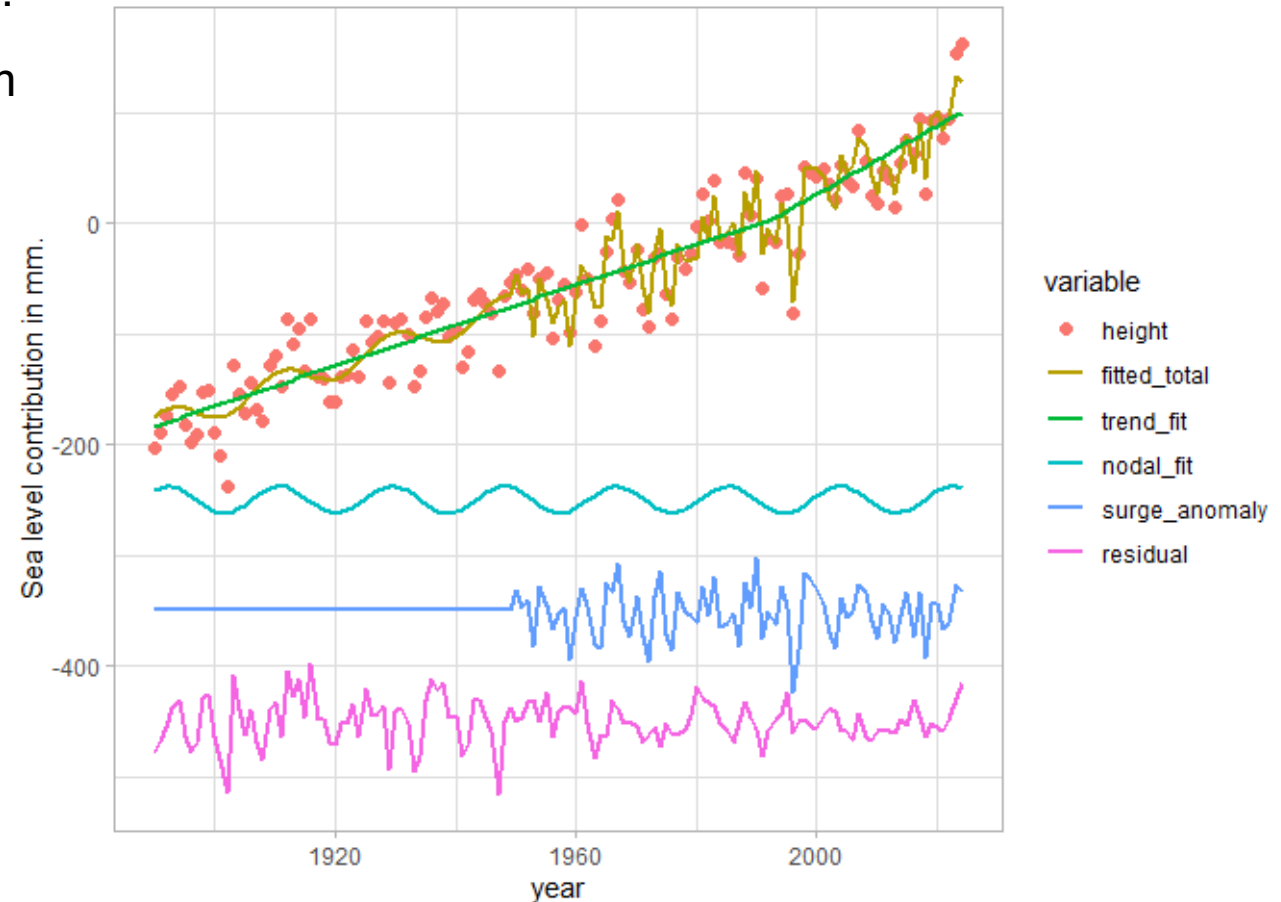


# Surge correction and Sea Level monitor

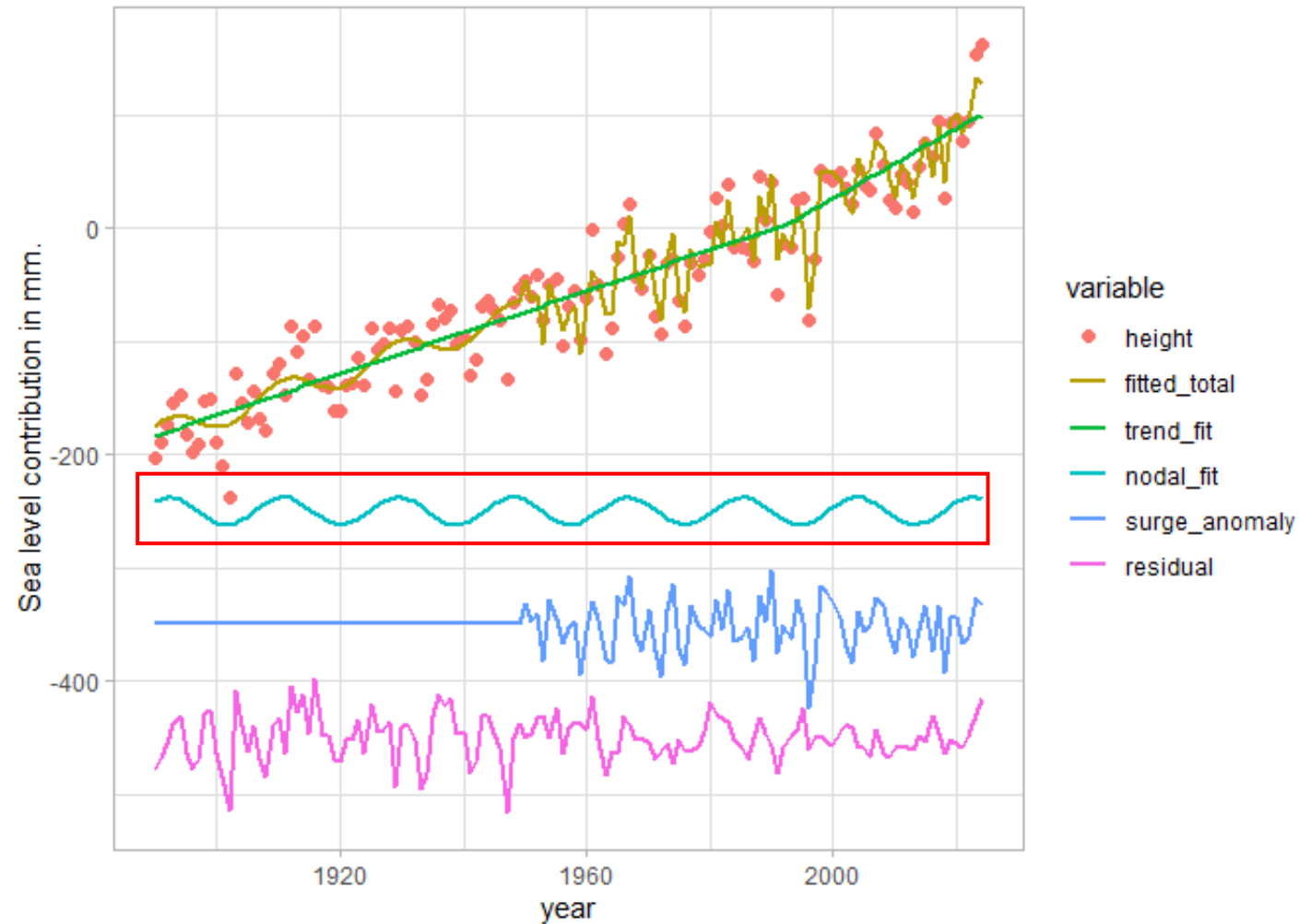
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Implementation of the surge correction for 1950-2025:

1. Yearly averages are calculated for 6 stations from the GTSM-ERA5 monthly means dataset
2. Surge anomaly is calculated as a deviation from mean per station and year
3. The surge anomalies are averaged across stations per year and used in the regression model as an offset



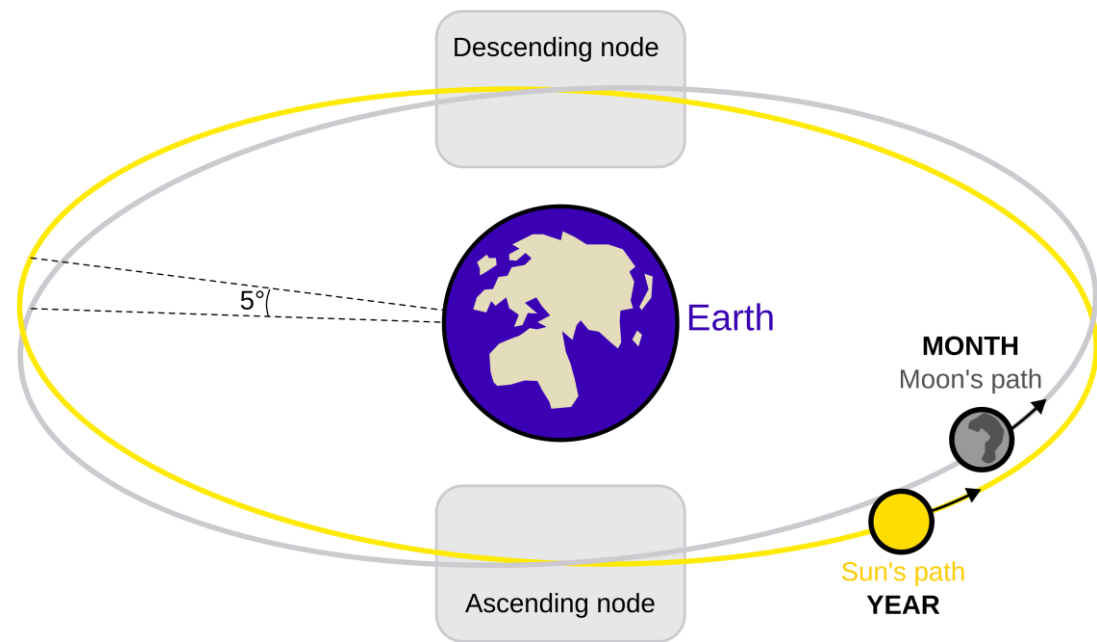
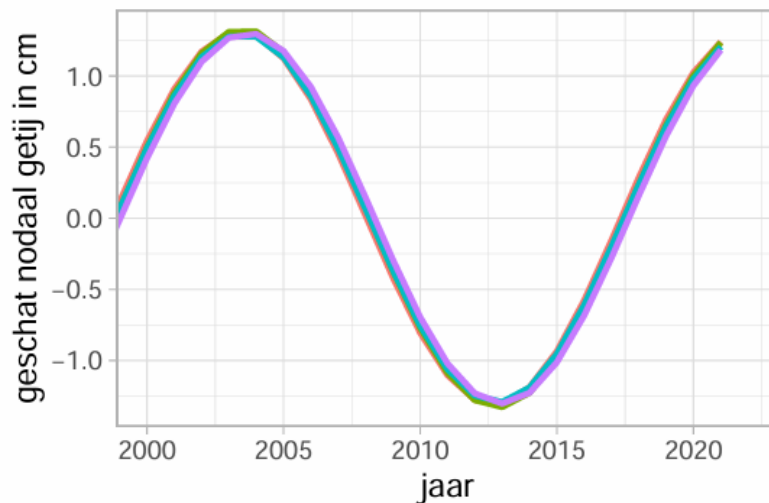
## 03 Nodal tides



## Nodal tide

- **Nodal tidal cycle** – long-period tide caused by astronomical effects (precession of the Moon's orbital plane relative to Earth's equatorial plane).
- Nodal tide is responsible for small year-to-year changes in sea level (~1-2cm amplitude over 18.6 year cycle)
- In Zeespiegelmonitor, the nodal tide is **estimated from observations** (as a regression term).

*Nodal tide amplitude  
estimate from observations*





## Nodal tide

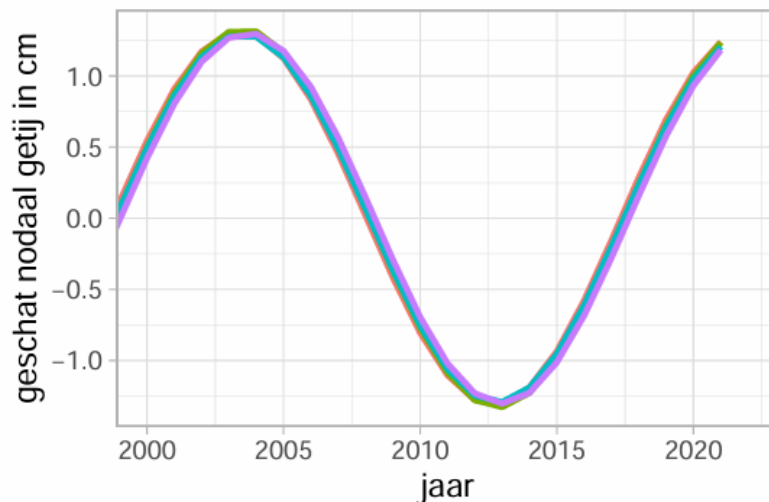
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- Nodal amplitude is theoretically described by the equilibrium tide, and is likely amplified by ocean mixing processes, i.e. long period changes in temperature and salinity (Bult et al., 2024).
- This process was investigated in more detail by tide gauge observations-derived nodal tide amplitude to GTSM (2D barotropic model) and GLORYS (3D ocean reanalysis with altimetry data assimilation).

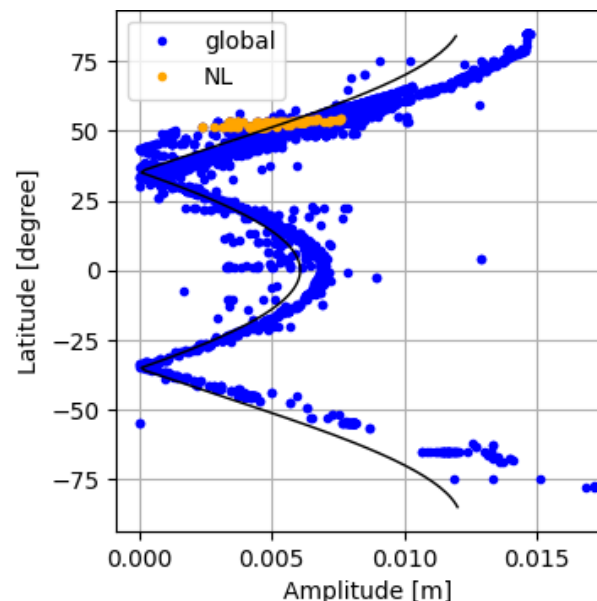
## Nodal tide

- Nodal tide in the global barotropic tide & surge model (GTSM) shows good agreement with the equilibrium tide formulation, estimating amplitude of ~3-7 mm at the Dutch coast.
- Analysis of the GLORYS water levels shows a larger amplitude of ~16 mm for Vlissingen. This model resolves 3D ocean currents, albeit at low coastal resolution, with data assimilation from altimetry, resulting a larger estimate closer to the estimate obtained in Zeespiegelmonitor from tide gauge analysis.

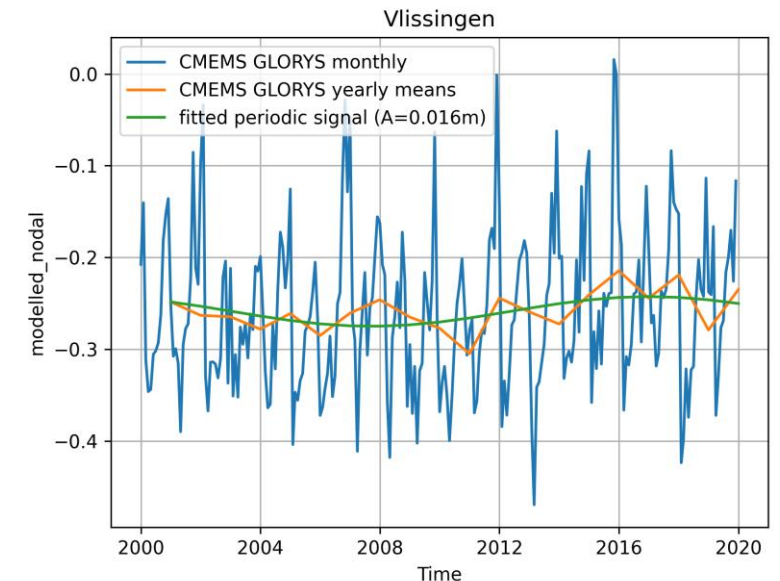
*Nodal tide amplitude  
estimate from observations*



*GTSM nodal tide vs. equilibrium tide:*



*Global Ocean Physics Reanalysis (GLORYS)*



# 04 Conclusions & Outlook

- Surge correction has been implemented starting from 1950 using ERA5
  - The computed surge compares well to measurements
  - The surge correction reduces variance significantly
  - Can it be extended further back?
- Nodal tides in North Sea are caused by:
  - Tidal potential, which is well understood and predictable, but
  - Ocean density and mixing, plausible but not modelled in a direct manner yet